Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/31/2005
PI Name:	Mazyck, David W. Ph.D.		
Project Title:	Magnetically Agitated Photocatalytic Reactor for Water Recovery		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32611-6450	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	9-CAN-00-BH
Start Date:	06/01/2002	**End Date:**	03/15/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chadik, Paul (University of Florida) Wu, CY (University of Florida) Powers, Kevin (University of Florida) Sigmund, Wolfgang (University of Florida) Meisel, Mark (University of Florida)		
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Task Description:	NASA has identified a number of water treatment options that have shown promise in space. The water recovery system on any particular mission will be a collection of individual treatment units, with each treatment unit focusing on a select group of contaminants. The goal of this research is to develop a post processor for the destruction of organic compounds and microbes that may not be removed in water subsystems prior to the MAPR. The overall objective is to develop a microgravity-compatible, compact reactor that is safe and reliable that provides product water that meets or exceeds NASA's potable water quality requirements. Furthermore, this innovative technology has a reduced ESM because of its compactness and ability to function reliably without the extensive use of consumables (i.e., infrequent replacement of the UV lamps or use of solar UV). The technology proposed herein is applicable for implementation on board the ISS, in-flight, or for planetary (e.g., Lunar, Mars, etc.) missions. The assessment phase of the project will quantify the reactor removal efficiency with respect to pH, conductivity, total organic carbon (TOC), and trace organics. Optimize the MAPR with respect to particle design and loading, number of UV lamps, and UV wavelength. Post processing of water from prior subsystems is crucial to ensure crew health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Magnetically Agitated Photocatalytic Reactor (MAPR) under development at the University of Florida is well suited for many water purification applications including recovering water during missions in space, removing contaminants from water in the home, providing drinking water onboard ocean vessels, supplying troops with drinking water during military operations, and providing safe water for disaster response when water is desperately required but not available. Additionally, the technology has homeland security applications for decontamination of intentionally poisoned water supplies and a safe method disposal of the chemical weapons arsenal. As sunlight can be used as the light source, this technique offers a unique and affordable solution to water quality problems that face developing countries. In addition to water applications, the MAPR could be designed for air purification applications including use in the airline industry, underground mining, or automobiles. Traditionally water has been purified of harmful contaminants by activated carbon adsorption or photocatalytic oxidation. Although both are effective, activated carbon has a finite adsorption capacity and requires replacement or regeneration. Photocatalytic slurries may encounter problems such as separation of photocatalysts from recovered water. The advantage of using magnetic substrates is that they can be agitated by an external magnetic field to provide enhanced mixing of the photocatalyst particles with pollutants, and to enhance exposure to UV light. In addition, because the magnetic substrates are controlled by a magnetic field, the recovered water will be free of the composite particles. Faster destruction rates allow for compact treatment devices, which are critical to the applications listed above. Furthermore, this innovative technology is attractive because of its compactness and ability to function reliably without the extensive use of consumables (i.e., very infrequent replacement of UV lamps and media).
Task Progress:	Recently, the catalyst synthesis process has been improved. A barium ferrite core that is commercially available rather than ball milled lead to a greater consistency in the catalyst performance. Double coatings of silica and titania respectively produced further improvements in performance such that a 10 mg/L phenol solution can be mineralized without photolysis in 7 hours with 3-365 nm lamps and 100 mg of catalyst. The magnetic field of the MAPR has been improved to increase the UV exposure of the catalyst. To accomplish this, a permanent magnet has been added above the reactor to counteract the force of gravity, which tends to cause the catalyst to remain near the bottom of the reactor. The result of this modification is that the reaction rate of mineralization is more than double that without the permanent magnet. In the context of long-term space missions, this is a highly advantageous modification as the field can be adapted to the gravitational requirements of the environment by using a DC electromagnet or changing permanent magnets. If a permanent magnet is used it will not contribute to increases in the power requirements for equivalent system mass calculation. The MAPR with the permanent magnet addition was optimized for catalyst mass, solenoid operating frequency and current using phenol as a model pollutant. The optimal catalyst mass was 400 mg with steep declines in performance if the mass is either increased or decreased. The phenol mineralization was relatively consistent for solenoid frequencies between 20 and 80 Hz above which performance declined significantly. Phenol mineralization increased linearly with current up to 1 A. Further increases in current did not improve performance, but increased the heat and power requirements.
Bibliography Type:	Description: (Last Updated: 10/04/2004)